BOUMEDIENE Allaa Eddine

Information Systems Engineer

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Professional Experience

CASPM *July 2026 – Present*
Information Systems Engineer *Mers El Hadjadj, Algeria*

- Business process modeling.
- Develop software with Node.js and Electron.js, including an attendance management system (replacing ZKTeco) and a stock management system.
- Create dynamic dashboards using Power BI.
- Maintain the company's software and databases and resolve day-to-day technical issues.

Kabima Fourniture *April 2026 – June 2026*
Information Systems Engineer *Oran, Algeria*

- Installed and configured equipment, networks and systems (e.g., NAS).
- Developed internal tools.
- Managed technical purchasing: sourcing, calls for tenders, IT and electrical equipment, quality control and follow-up until delivery.

Giplait Laiterie La SOURCE *March 2025 – April 2025*
Intern – Management Information Systems Department *Saida, Algeria*

- Designed a Decision Support System (DSS) to optimize production, using metaheuristic algorithms, notably Teaching-Learning-Based Optimization (TLBO) and the Tiki-Taka Algorithm (TTA).

Algerian Cement Industrial Group (GICA) *April 2024 – May 2024*
Intern – Management Information Systems Department *Saida, Algeria*

- Modeled corrective maintenance processes with BPMN and managed IT assets with GLPI.

CHIALI SERVICES *January 2024 – February 2024*
Intern – IT Solutions Department *Sidi Bel Abbes, Algeria*

- Designed UML models and developed a server monitoring application.

Algerian Cement Industrial Group (GICA) *March 2023 – April 2023*
Intern – Management Information Systems Department *Saida, Algeria*

- Developed interactive HR dashboards in Power BI using a star schema.

Sonatrach (GP1Z Zone) *December 2022 – January 2023*
Intern – IT and Management Department *Oran, Algeria*

- Explored industrial networks and administered Oracle databases.

Education

State Engineering Degree & Master's Degree in Information Systems Engineering and Management — *National Polytechnic School of Oran – Maurice Audin*
Oran, Algeria — *2022 – 2025*
Final-year thesis: Optimization of metaheuristic algorithms and their applications in various domains

Preparatory Cycle in Science and Technology — *Higher School of Electrical and Energy Engineering of Oran*
Oran, Algeria — *2020 – 2022*
Preparatory training for engineering studies

Technical Skills

Development	Java, Express.js, Electron.js, Vue.js, Tailwind CSS, Bootstrap, Git
Databases	SQL (PostgreSQL, MySQL, Oracle), NoSQL (MongoDB)
Data & BI	Data cleaning with Python (e.g., Pandas); Power BI (cleaning and visualization), dimensional modeling
Data Science	Time series (ARIMA, SARIMA), XGBoost, U-Net
Management & IS	BPMN, UML, GLPI, Odoo, MS Project
Infrastructure	Network configuration, operating systems, IT systems administration
Certifications	ITIL v4, CCNA 1, Data Analytics

Languages

Arabic	Native
English	Advanced
French	Intermediate

Projects

Development

Attendance Management System
Attendance tracking application (Electron.js, Node.js, MySQL) replacing the ZKTeco system.

El Ktawther
Student management application (Electron.js, SQLite): registration, search and student ID cards.

Power BI

Job Application Tracking
Data preparation (cleaning, normalization), star schema and interactive dashboard for job application analysis.

Data Science

Skin Lesion Segmentation
U-Net with decision-threshold optimization using the ITTA-TLBO metaheuristic algorithms (Dice score: 0.888).

Pharmacy Sales Forecasting
Comparative analysis of XGBoost and SARIMA models on time series.